Worksheet 4.2 — Measures of Central Tendency and Spread

Mathematics: Analysis & Approaches SL (IB) · Unit 4: Statistics and Probability

The Big Ideas

Standard deviation, quartiles, and the outlier rule describe how spread out data is.

- $\sigma = \sqrt{\dfrac{\sum(x-\bar{x})^2}{n}}$.
- Outlier rule: below $Q_1 - 1.5\,IQR$ or above $Q_3 + 1.5\,IQR$.
- Adding a constant shifts the mean, not the SD; multiplying scales both.

Worked Examples

Example 1 — Mean and median

Find the mean and median of $6, 6, 11, 14, 23$.

Solution. Mean $= \dfrac{60}{5} = 12$; median (middle value) $= 11$.

Conclusion: mean 12, median 11.

Example 2 — Standard deviation

Find the SD of $4, 6, 8, 10, 12$.

Solution. Mean $= 8$; deviations squared: $16, 4, 0, 4, 16$; sum $= 40$; $\sigma = \sqrt{8}$.

Conclusion: $\sigma \approx 2.83$.

Example 3 — Outlier check

A data set has $Q_1 = 22$, $Q_3 = 38$. Is 65 an outlier?

Solution. $IQR = 16$; upper fence $= 38 + 24 = 62$.

Conclusion: yes, $65 > 62$.

Example 4 — Transformation

A data set has mean 55, SD 6. Every value is tripled, then 4 is subtracted. Find the new mean and SD.

Solution. Tripling: mean 165, SD 18; subtracting 4 changes only the mean.

Conclusion: new mean 161, SD 18.

Example 5 — Lower outlier check

A data set has $Q_1 = 30$, $Q_3 = 50$. Is 2 an outlier?

Solution. $IQR = 20$; lower fence $= 30 - 30 = 0$.

Conclusion: no, since $2 > 0$ (it's above the lower fence).

Example 6 — Finding a missing data value from the mean

Five values have a mean of 14: $8, 12, 17, 19, x$. Find x.

Solution. Sum must be $5(14) = 70$. $8 + 12 + 17 + 19 = 56$, so $x = 70 - 56$.

Conclusion: $x = 14$.

Example 7 — SD from a frequency table

Find the SD of the data 2(freq 2), 4(freq 3), 6(freq 1) (6 values total).

Solution. Mean $= \dfrac{2(2) + 4(3) + 6(1)}{6} = \dfrac{4 + 12 + 6}{6} = \dfrac{22}{6} \approx 3.67$.

Deviations squared (weighted): $2(2-3.67)^2 + 3(4-3.67)^2 + 1(6-3.67)^2 \approx 2(2.79) + 3(0.109) + 1(5.43) \approx 11.14$.

Conclusion: $\sigma = \sqrt{11.14/6} \approx 1.36$.

Example 8 — Comparing spread using range vs IQR

Data set E has range 40 but IQR 5; data set F has range 20 but IQR 12. Which is more consistent in its middle 50%, and why might the ranges be misleading?

Solution. IQR measures the middle 50% spread, ignoring extreme values; E's small IQR (5) shows most data is tightly clustered despite one or two extreme outliers stretching the range.

Conclusion: E is more consistent in its middle half; its large range is likely driven by outliers, which the IQR correctly ignores.

Example 9 — Application — effect of a scaling error

A scale reads every mass 2 kg too high, and additionally every reading was mistakenly recorded in a unit requiring multiplication by 1.1. If the true mean is 60 kg and true SD is 4 kg, find the reported mean and SD (assume the $+2$ happened before the $\times 1.1$).

Solution. Add 2: mean 62, SD unchanged 4. Multiply by 1.1: mean $62(1.1) = 68.2$, SD $4(1.1) = 4.4$.

Conclusion: reported mean 68.2 kg, SD 4.4 kg.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Find the mean and median of $2, 5, 9, 12, 27$.

Question 2

Find the SD of $3, 7, 11, 15, 19$.

Question 3

A data set has $Q_1 = 10$, $Q_3 = 20$. Is 36 an outlier?

Question 4

A data set has mean 30, SD 4. Every value has 12 added. Find the new mean and SD.

Question 5

A data set has mean 20, SD 3. Every value is doubled. Find the new mean and SD.

Question 6

Class A: median 80, IQR 6. Class B: median 76, IQR 18. Compare performance and consistency.

Question 7 — Challenge

A data set has five-number summary $5, 15, 20, 28, 55$. Determine whether 55 is an outlier.

Question 8 — Challenge

A data set has mean 44 and SD 5. Every value is halved and then 10 is added. Find the new mean and SD.

Question 9

A data set has $Q_1 = 25$, $Q_3 = 45$. Is the value 1 an outlier?

Question 10

Five values have a mean of 20: $14, 18, 22, 25, x$. Find x.

Question 11 — Challenge

Find the SD of the data 3(freq 1), 5(freq 4), 7(freq 1) (6 values total).

Question 12 — Challenge

Data set G has range 50 but IQR 6; data set H has range 15 but IQR 10. Which is more consistent in its middle 50%, and explain why the ranges alone would be misleading.

Question 13 — Challenge

A sensor overreads every value by 3 units, and its readings must then be multiplied by 1.2 to correct the unit. If the true mean is 50 and true SD is 6, find the reported mean and SD.

Question Recap & Answer Key

Question Recap

1. Find the mean and median of $2, 5, 9, 12, 27$.
2. Find the SD of $3, 7, 11, 15, 19$.
3. A data set has $Q_1 = 10$, $Q_3 = 20$. Is 36 an outlier?
4. A data set has mean 30, SD 4. Every value has 12 added. Find the new mean and SD.
5. A data set has mean 20, SD 3. Every value is doubled. Find the new mean and SD.
6. Class A: median 80, IQR 6. Class B: median 76, IQR 18. Compare performance and consistency.
7. A data set has five-number summary $5, 15, 20, 28, 55$. Determine whether 55 is an outlier.
8. A data set has mean 44 and SD 5. Every value is halved and then 10 is added. Find the new mean and SD.
9. A data set has $Q_1 = 25$, $Q_3 = 45$. Is the value 1 an outlier?
10. Five values have a mean of 20: $14, 18, 22, 25, x$. Find x.
11. Find the SD of the data 3(freq 1), 5(freq 4), 7(freq 1) (6 values total).
12. Data set G has range 50 but IQR 6; data set H has range 15 but IQR 10. Which is more consistent in its middle 50%, and explain why the ranges alone would be misleading.
13. A sensor overreads every value by 3 units, and its readings must then be multiplied by 1.2 to correct the unit. If the true mean is 50 and true SD is 6, find the reported mean and SD.

✓ Answer Key

1. mean = 11, median = 9
2. mean = 11; deviations sq: $64, 16, 0, 16, 64 = 160$; $\sigma = \sqrt{32} \approx 5.66$
3. $IQR = 10$, fence $= 20 + 15 = 35$; $36 > 35$: yes, outlier
4. mean 42, SD unchanged 4
5. mean 40, SD 6
6. A has higher median (better typical score) and smaller IQR (more consistent)
7. $IQR = 13$, fence $= 28 + 19.5 = 47.5$; $55 > 47.5$: yes, outlier
8. halving: mean 22, SD 2.5; +10 changes only mean: mean 32, SD 2.5
9. $IQR = 20$, lower fence $= 25 - 30 = -5$; $1 > -5$: not an outlier
10. sum must be 100; $14 + 18 + 22 + 25 = 79$, so $x = 21$
11. mean $= \frac{3+20+7}{6} = 5$; deviations sq (weighted): $1(4) + 4(0) + 1(4) = 8$; $\sigma = \sqrt{8/6} \approx 1.15$
12. H is more consistent in its middle half (smaller IQR); G's large range is likely driven by one or two extreme outliers, which the IQR ignores
13. add 3: mean 53, SD unchanged 6; multiply by 1.2: mean 63.6, SD 7.2